

MANIKANDAN GANESAN

Senior Principal Engineer

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Professional Summary

Senior Principal Engineer with **13+ years of experience** designing and building large-scale distributed systems, cloud-native platforms, and backend services supporting **millions of users**. Hands-on technical leader with deep experience in software architecture, microservices, multi-tenant systems, workflow orchestration, security, scalability, and production reliability.

Architect and key contributor to enterprise **Backup, Security & Compliance platforms**, including distributed workflow frameworks, cloud data-processing services, multi-tenant orchestration systems, and reusable platform capabilities. Experienced in modernizing legacy applications into scalable microservices architectures and designing systems for high availability, performance, security, and operational efficiency. Strong hands-on expertise in **Node.js, TypeScript, Go, PHP, AWS, Docker, PostgreSQL, Microsoft SQL Server, Redis, DynamoDB, and event-driven architectures**.

Experienced in architecture and code reviews, CI/CD, observability, performance optimization, incident response, technical prototyping, and mentoring engineers.

Proven ability to lead complex engineering initiatives from architecture and design through implementation, deployment, and production operations while influencing technical direction across engineering teams.

Skills

TECHNICAL SKILLS

Cloud & Infrastructure: AWS — EC2, ECS, Fargate, Auto Scaling, S3, SQS, Lambda, RDS, DynamoDB, FSx for Lustre, KMS, IAM, STS, Secrets Manager, CloudWatch, CodeBuild, CodeDeploy, CodePipeline;

Programming & Backend: Node.js, TypeScript, JavaScript, Go, PHP, NestJS, SQL

Architecture & Distributed Systems:

Microservices, Distributed Systems, Cloud-Native Architecture, Event-Driven Architecture, Multi-Tenant Systems, Workflow Orchestration, Asynchronous Processing, Retry & Recovery Patterns, High Availability, Scalability

Databases & Caching: PostgreSQL, Microsoft SQL Server, Redis, DynamoDB, Elasticsearch

DevOps & Observability: Docker, Kubernetes, Linux, Git, CI/CD, Infrastructure as Code, PM2, Grafana, Kibana, CloudWatch, SonarQube

Security Engineering: Multi-Tenant Isolation, Least-Privilege Access, IAM/STS Credential Management, KMS Envelope Encryption, Secrets Management, Client-Side Encryption, Secure Containerized Workloads

Engineering Leadership: Software Architecture, System Design, Architecture Reviews, Code Reviews, Technical Mentoring, Production Incident Response, Performance Optimization, Cost Optimization, Release Engineering

Professional Experience

Senior Principal Engineer

SysCloud, Inc

Sep 2023 - Current

Atlanta, United States

- Lead architecture and technical design for **distributed, cloud-native platforms** supporting backup, restore, data processing, security, and compliance workloads across large-scale multi-tenant environments.
- Co-designed and developed a **Multi-Tenant Orchestrator**, enabling tenant-aware workload execution, workload isolation, dynamic resource allocation, reconciliation, and automated recovery across distributed processing environments.
- Drive architecture and design decisions across **Node.js, TypeScript, Go, AWS, Docker, PostgreSQL, Redis, DynamoDB, SQS, S3, FSx for Lustre, IAM, STS, and KMS**, balancing scalability, reliability, security, performance, and infrastructure cost.
- Lead the architecture and development of **Cloud Mover microservices**, providing reusable distributed workflows for backup, restore, export, archive processing, data movement, and large-scale asynchronous workloads.
- Designed security architecture for multi-tenant workloads using **least-privilege IAM, temporary STS credentials, KMS-based encryption, secure credential lifecycle management, tenant isolation, and container/process-level security controls**.
- Delivered key platform security capabilities including **client-side encryption and ransomware protection**, strengthening data protection and recovery capabilities.
- Established engineering standards across source control and release processes by implementing a **company-wide Git workflow**, improving development consistency, collaboration, and release management.
- Evaluate architecture and infrastructure options across AWS services and deployment models, considering scalability, reliability, operational complexity, performance, security, and cost.
- Provide technical leadership through system design reviews, architecture discussions, mentoring, production troubleshooting, and incident resolution while remaining actively involved in hands-on engineering.

Principal Engineer

Syscloud Technologies

Aug 2018 - Aug 2023

Chennai, India

- Led architecture and development of **cloud-native backend and microservices platforms**, owning major engineering initiatives from requirements and system design through implementation, deployment, monitoring, and production support.
- Evolved **Cloud Mover and distributed processing services** into reusable platform capabilities supporting backup, restore, data movement, export, archive processing, and other large-scale asynchronous workflows.
- Designed and deployed scalable microservices on **AWS ECS/Fargate**, using services including Redis, SQS, DynamoDB, PostgreSQL, S3, Secrets Manager, and CloudWatch to support reliable distributed processing.
- Defined service boundaries, APIs, reusable components, and engineering patterns across microservices, improving maintainability and enabling teams to deliver new capabilities without duplicating common platform functionality.
- Improved distributed workflow reliability by establishing consistent approaches for **error handling, retries, failure recovery, timeouts, state management, and operational monitoring** across backend services.
- Introduced architectural and engineering patterns for cross-cutting concerns such as **logging, security, caching, and configuration management**, improving consistency and maintainability across services.
- Strengthened application performance and security through **Redis-based caching, centralized secrets management, infrastructure configuration, and cloud resource optimization**.
- Established and reinforced engineering practices including **code reviews, automated testing, linting, build automation, CI/CD, deployment validation, and production readiness**, improving release consistency and software quality.
- Provided technical leadership for an **8-member engineering team**, leading design discussions, reviewing implementations, mentoring engineers, planning technical work, and supporting delivery of complex platform initiatives.
- Partnered with engineering and product stakeholders to investigate production issues, evaluate architectural tradeoffs, and translate business requirements into scalable technical solutions.

Lead Engineer

Syscloud Technologies

Aug 2017 - Aug 2018

Chennai, India

- Led the migration of a **legacy monolithic application to a microservices architecture**, improving scalability, maintainability, deployment flexibility, and service-level ownership.
- Designed and developed a **reusable Node.js workflow framework** that standardized common execution patterns across microservices, including task sequencing, shared utilities, workflow state handling, and reusable processing logic.
- Built reusable backend components and shared libraries that reduced duplication across services and accelerated delivery of new product capabilities.

- Drove the transition from traditional server deployments to **containerized, automated deployments using Docker and Git-based CI/CD pipelines**, improving deployment consistency and release reliability.
- Implemented AWS deployment automation and infrastructure patterns for scalable backend workloads, including **Auto Scaling Groups, centralized configuration management, and cloud-based monitoring**.
- Migrated application configuration to **AWS Systems Manager Parameter Store**, improving centralized management, security, and operational control across environments.
- Improved application scalability by modernizing workloads for **Linux-based cloud infrastructure** and implementing dynamic scaling strategies for variable workloads.
- Strengthened production reliability through infrastructure monitoring, proactive issue detection, and performance tuning across application and cloud resources.
- Conducted architecture and code reviews, optimized backend implementations, and mentored engineers on software engineering and CI/CD practices, contributing to a **20% faster deployment cycle**.
- Containerized high-throughput processing workloads and migrated supporting services and databases to scalable cloud infrastructure to improve performance under increased load.

Senior Software Engineer

Aug 2015 - Aug 2017

Syscloud Technologies

Chennai, India

- Played a key engineering role in building and evolving the company's **Data Loss Prevention (DLP) and Compliance platform**, delivering backend capabilities for security monitoring, policy enforcement, classification, and reporting.
- Developed product capabilities for detecting and classifying sensitive or risky content across customer environments, supporting policies related to **PCI, FERPA, HIPAA, PII, cyberbullying, life-threat indicators, and objectionable content**.
- Built and enhanced backend workflows for **content extraction, policy evaluation, classification, threshold handling, exception processing, and reporting**, improving the platform's ability to process and surface security and compliance events.
- Worked closely with Sales and customer-facing teams to understand real-world security and compliance requirements, translating customer feedback into product enhancements and new backend capabilities.
- Contributed to real-time processing workflows that responded to changes in files, emails, attachments, users, and other cloud objects, enabling timely backup and security scanning.
- Improved application scalability and operational efficiency by containerizing workloads and deploying services on **AWS ECS/EC2 and Fargate**.
- Tuned AWS scaling policies and monitored application performance across services to maintain responsiveness and reliability as workload volume increased.
- Helped establish stronger environment isolation and cloud deployment practices using **Docker and AWS**, contributing to the product's transition toward more scalable, independently deployable services.

Software Engineer

Jun 2013 - Aug 2015

Syscloud Technologies

Chennai, India

- Developed backend features for the company's backup and security platform using **PHP and Microsoft SQL Server**, contributing to core product functionality and production reliability.
- Designed and optimized database components including **schemas, stored procedures, triggers, indexes, and SQL queries** to improve performance and support growing application workloads.
- Built and delivered **Real-Time Backup** capabilities, adding important product functionality for timely data protection and recovery.
- Worked with senior engineers to troubleshoot customer issues, analyze production behavior, and implement reliable technical fixes.
- Collaborated with Sales and customer-facing teams to understand user requirements and translate feedback into backend product enhancements.
- Contributed to the early evolution of the platform's backend and infrastructure, building a strong foundation in scalable application design, production support, and cloud-based systems.

Education

Bachelor of Engineering: Computer Science

Annamalai University

Jun 2007 - Mar 2011

Chidambaram, Tamilnadu, India